12-26 第2篇

阅读理解（说明文） Why Does a Rainbow Have Exactly Seven Colours?

① Almost every middle school student in China has, at some point after a summer shower, looked up and seen a **rainbow arching across the sky** — a gentle half-circle of **red, orange, yellow, green, blue, indigo and violet**. Many of us have also asked a quiet question: **why exactly seven colours? Why not five? Why not nine?** The full answer is more interesting than most people think. In fact, a rainbow is not "seven" colours from nature; it is **millions of colours blended smoothly together**, and the number *seven* comes partly from physics — and partly from the decision of one scientist in England more than 350 years ago. This short article will explain, in simple words, the four small facts you need to understand a rainbow.

② **Fact 1 — Sunlight only looks white.** When you stand in a sunny classroom, the sunlight pouring through the window appears to be a single clear colour we casually call "white". In truth, that white light is a **mixture of every colour the human eye can see**, travelling together in a tidy invisible bundle. You can prove this at home with one tool your school physics lab has in every drawer: a **glass prism**. Hold a triangular prism in a thin sunbeam, and on the wall opposite you will suddenly see a **bright small strip of colour** — the rainbow, in miniature. Isaac Newton did exactly this experiment in 1666, in a darkened room with a small hole in the window shutter, and he was the first person to prove, with careful measurements, that white sunlight is actually a bundle of many colours held together.

③ **Fact 2 — A single raindrop is a tiny round "prism".** After a shower, the sky is filled with millions of tiny **spherical (ball-shaped) raindrops**, each smaller than a grain of rice. When one sunbeam enters one raindrop, three small things happen, one after another. First, the sunbeam **bends a tiny bit** as it passes from air into water — this is called **refraction**. Second, the light hits the inside back wall of the raindrop and **bounces once** — this is called **reflection**. Third, the light **bends again** as it leaves the raindrop and returns to the air. These three gentle actions — bend, bounce, bend — happen, together, inside one raindrop in less than **one-millionth of a second**. And because different colours of light bend by **slightly different amounts** (red bends least, violet bends most), the light that finally exits the raindrop is **no longer a tight bundle**: it has been gently **spread out into a fan** of colours. A rainbow is simply the **combined fan of colours** from millions of raindrops, all spraying the same spread pattern across the sky at the same moment.

④ **Fact 3 — The order never changes.** In every rainbow, large or small, anywhere on Earth, the colours always appear in the **same strict order** — red on the outside of the

arch, violet on the inside — because the bending angle of each colour of light is a **fixed physical number** in water. **Red** light bends at about **42°** from the direction of the sun; **violet** light bends at about **40°**. In between these two angles sit every other colour of the visible fan. This is why you will never see a rainbow with violet on the outside, or with the colours scrambled — such a rainbow would violate the simple physics of refraction, and nature does not violate itself. **A rainbow, in other words, is one of the most trustworthy sights in nature: its order is the same in Beijing, in Paris, and on a small country road in Kenya.**

⑤ **Fact 4 — The number "seven" is partly a choice.** Here is the surprise. A real rainbow contains **not seven colours but an unbroken smooth spectrum**, with each colour gliding gently into the next. There is no real line between "yellow" and "green"; they simply blend. So why do we say "seven"? Isaac Newton, when he studied the rainbow carefully, first counted only **five bands**: red, yellow, green, blue, violet. Later, because he believed the rainbow should match **the seven notes of a musical scale** (a popular idea in Europe in the 1660s), he **added two more colours** — orange (between red and yellow) and indigo (between blue and violet) — to bring the count to **seven**. Different cultures in history have counted differently: some African traditions describe three bands, some early Chinese poems describe five, and several Native American peoples describe six. **The physics does not change; only the naming does.** When you look up at a rainbow today, remember: nature is giving you a smooth fan, and human culture is dividing it, gently, into seven names.

⑥ A last small practical fact: to see a rainbow, three conditions must hold at the same time. The **sun must be behind you**; there must be **falling raindrops in front of you**; and the **angle between your eye, the sun, and the raindrops** must be close to 42°. That is why rainbows are more common in the late afternoon, **after a shower has moved east while the setting sun is behind you in the west**. If you remember this small rule, you may catch more rainbows in a single summer than most adults catch in a whole decade.

⑦ So the next time a classmate asks, *"Why does a rainbow have seven colours?"*, you can gently answer: it has **millions of colours** from nature, arranged in **one unchanging physical order** by the bending of light inside raindrops, and "seven" is a **gentle human naming** chosen in 1666 to match the seven notes of a musical scale. A rainbow, then, is where **physics and poetry quietly meet** in the sky — and middle-school students happen to be in the perfect age to notice both.

注释： rainbow 彩虹；prism 棱镜；spherical 球形的；refraction 折射；reflection 反射；spectrum 光谱；visible 可见的

根据短文内容，选择最佳答案。

1. What is Fact 1 about sunlight, and how can it be proven in a classroom?

A. Sunlight is truly one single colour called white, and no experiment can change that fact.
B. When one stands in a sunny classroom, the sunlight pouring through the window appears to be a single clear colour casually called "white"; in truth that white light is a mixture of every colour the human eye can see, travelling together in a tidy invisible bundle — which can be proven at home with one tool every school physics lab keeps in a drawer, a glass prism: hold a triangular prism in a thin sunbeam and on the opposite wall a bright small strip of colour suddenly appears — the rainbow in miniature — exactly as Isaac Newton did in 1666 in a darkened room with a small hole in the window shutter, the first person to prove with careful measurements that white sunlight is actually a bundle of many colours held together.
C. Sunlight has only three colours and can be seen without any tool.
D. Only satellites in space can measure the colours inside sunlight.

2. What three small actions happen inside a single raindrop, according to Fact 2?

A. The light explodes loudly inside the raindrop, destroying it.
B. The light passes through the raindrop without any change at all.
C. When one sunbeam enters one raindrop, three small things happen one after another: first, the sunbeam bends a tiny bit as it passes from air into water (refraction); second, the light hits the inside back wall of the raindrop and bounces once (reflection); third, the light bends again as it leaves the raindrop and returns to the air — these three gentle bend-bounce-bend actions happen together in less than one-millionth of a second, and because different colours bend by slightly different amounts (red bends least, violet bends most), the light finally exits as a spread-out fan of colours; a rainbow is simply the combined fan from millions of raindrops spraying the same pattern at the same moment.
D. The light freezes inside the raindrop and stays trapped there forever.

3. Why does every rainbow on Earth have the same colour order, according to Fact 3?

A. Because humans paint the same order onto every rainbow by habit.
B. In every rainbow, large or small, anywhere on Earth, the colours always appear in the same strict order — red on the outside of the arch, violet on the inside — because the bending angle of each colour of light is a fixed physical number in water: red bends at about 42° from the direction of the sun and violet at about 40°, with every other visible colour fitting between them; that is why one will never see a rainbow with violet on the outside or with scrambled colours, such a rainbow would violate the simple physics of refraction, and nature does not violate itself — making a rainbow one of the most trustworthy sights in nature, with the same order in Beijing, in Paris, and on a small country road in Kenya.
C. Different rainbows on Earth have completely different colour orders.
D. The order depends on the mood of the person looking.

4. What does Fact 4 say about the number "seven" in rainbows?

A. A real rainbow is strictly seven separate lines drawn by nature with visible borders between each colour.
B. Rainbows actually have 24 hard-bordered colours and no human has ever miscounted.
C. A real rainbow contains not seven colours but an unbroken smooth spectrum with each colour gliding gently into the next (no real line between yellow and green — they simply blend); Newton first counted only five bands (red, yellow, green, blue, violet), but because he believed the rainbow should match the seven notes of a musical scale — a popular idea in Europe in the 1660s — he added orange (between red and yellow) and indigo (between blue and violet) to bring the count to seven, while different cultures counted differently (some African traditions three bands, some early Chinese poems five, several Native American peoples six); the physics does not change, only the naming does.
D. Rainbows always have exactly seven colours in every culture and no human has ever counted differently.

5. According to paragraph ⑥, what three conditions must hold for a rainbow to be visible?

A. The sun must be directly overhead, the sky must be perfectly dry, and the viewer must be indoors.
B. The sun must be behind you, there must be falling raindrops in front of you, and the angle between your eye, the sun, and the raindrops must be close to 42°; this is why rainbows are more common in the late afternoon, after a shower has moved east while the setting sun is behind you in the west, and keeping this small rule in mind may let one catch more rainbows in a single summer than most adults catch in a whole decade.
C. There must be snow and a cloudless sky at the same time.
D. Rainbows are visible only at midnight under strong moonlight.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段
【解析】"白光=多色束 / 三棱镜 / 牛顿 1666 暗室小孔"直引。A、C、D 均错。故选 B。

2.
【答案】C
【题型】细节理解题
【定位】第③段
【解析】"折射 + 反射 + 折射 / 红折最小 紫折最大 / 组合扇"直引。A、B、D 均错。故选 C。

3.
【答案】B
【题型】细节理解题
【定位】第④段
【解析】"红 42° 紫 40° / 物理固定 / 全球同序"直引。A、C、D 均错。故选 B。

4.
【答案】C
【题型】细节理解题
【定位】第⑤段
【解析】"本无七色 / 牛顿原 5 色 / 7 音阶加 orange/indigo / 非洲 3 / 中国 5 / 北美 6"直引。A、B、D 均错。故选 C。

5.
【答案】B
【题型】细节理解题
【定位】第⑥段
【解析】"背阳 / 前雨 / 42°角 / 午后常见"直引。A、C、D 均错。故选 B。